

Special Notice

Request for Information:

Flexible and Malleable Battery Test and Evaluation Support

DARPA-SN-25-64

March 24, 2025



Defense Advanced Research Projects Agency

Microsystems Technology Office

675 North Randolph Street

Arlington, VA 22203-2114

**Request for Information (RFI)
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Flexible and Malleable Battery Test and Evaluation Support RFI

**Defense Advanced Research Projects Agency (DARPA)
Microsystems Technology Office (MTO)**

Posting Date: March 24, 2025

Responses Due: April 21, 2025, at 5:00 p.m. Eastern Time (ET)

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RFI DESCRIPTION:

The Defense Advanced Research Projects Agency (DARPA) Microsystems Technology Office (MTO) seeks information on independent verification and validation (IV&V) capabilities for testing and evaluation of flexible or malleable batteries. Flexible and malleable batteries present IV&V challenges that are not found with current rigid designs, such as the need for a combination of mechanical testing with battery discharge monitoring.

The goal of this Request for Information (RFI) is feedback regarding the testing, evaluation, and metrology of flexible and/or malleable battery systems from stakeholders including industry, university-affiliated research centers (UARCs), government labs, and federally funded research and development centers (FFRDCs) detailing infrastructure, equipment, testing methodology, and expertise related to the validation of battery systems with flexible or malleable characteristics. Responses to this RFI will be used to inform and support future DARPA solicitations on related topics.

BACKGROUND:

In recent decades, research into battery systems has expanded beyond the scope of standard rigid-packaged battery configurations into those with soft, flexible, malleable, or self-healing properties. Flexible batteries are defined here as batteries that can support the storage and release of energy during bending, whereas malleable batteries are those batteries that can be deformed in one or multiple directions and still sustain storage and release of energy. The use case of these batteries is directly coupled with a mechanical function that includes either bending or deformation.

Flexible and malleable batteries offer a path to fundamentally reduce the physical and cognitive load on the warfighter by reducing carry weight while increasing the number of potential uses for an individual battery beyond that of traditional cylindrical or prismatic batteries. Shape-changing batteries could also be inserted into traditionally-unusable voids in devices, which would change how we utilize space and reduce overall volume. Alternative production methods and materials may further change the scale at which batteries must be manufactured, moving these facilities closer to the user and guaranteeing reliable access to power even when traditional battery supplies are disrupted.

There are numerous examples of batteries reported to operate during bending or deformation. However, it remains unclear how to systematically evaluate these systems in a way that probes the coupling between key mechanical and electrochemical functions. For example: if a battery is deformed how does the electrochemical performance, including state of charge (SOC), energy and power density, nominal cell voltage, etc., respond during and after the mechanical deformation? Furthermore, how would such measurements be used to inform improvement or advancement of the technology?

The key unknowns in testing these devices relate to *testing capabilities*, e.g., what tools and capabilities currently exist for controlled electrochemical-mechanical testing of batteries, as well as *testing methodology*, e.g., what tests are most effective to identify the coupling between the mechanical and electrochemical device functions.

Finally, as battery systems incorporate multiple interconnected layers of active and inactive materials, the mechanisms associated with mechanical-electrochemical coupling are associated with the changing properties of the materials and interfaces under stress. In such cases, metrology of the materials and interfaces remains important to interpret and validate the impact of mechanical stresses on the electrochemical performance (and vice versa). For example: reversible changes may occur, such as the change of resistance in an electrolyte under elastic strain. Alternatively, irreversible changes may occur, such as the formation of a crack or void at the electrode-electrolyte interface. For flexible and malleable batteries, what metrology methods are most suitable to characterize, understand, and validate performance targets?

REQUESTED INFORMATION:

DARPA seeks innovative insights to address the technical challenge area(s) described below. Responses are welcome from all capable sources including, but not limited to, private or public companies, individuals, universities, university-affiliated research centers, not-for-profit research institutions, and U.S. Government-sponsored labs. Responses to topic 1 below are of greatest interest for this RFI, but responses to all topics below are welcome.

Topic 1: Mechanical–Electrochemical Testing of Batteries

- Describe IV&V infrastructure that currently exists to test and evaluate flexible or malleable battery systems.
 - What facilities, tools, and equipment are available that have been used in the past for mechanical–electrochemical testing?
 - What facilities, tools, and equipment are available which could be combined and used for the purpose of mechanical–electrochemical tests?
- Describe mechanical-electrochemical testing methodology for flexible or malleable battery systems.
 - What techniques reflect best practices for mechanical–electrochemical testing of flexible and malleable batteries?
 - How can battery characteristics such as SOC, cell resistance, nominal cell voltage, energy density, etc. be analyzed during and after bending and/or deformation of a battery?
 - What validation metrics derived from testing are best suited to inform and evaluate improvement of the combined mechanical and electrochemical system functions?
- Describe team expertise in mechanical-electrochemical testing.
 - What previous experience do team members possess in electrochemical and mechanical testing of batteries?

Topic 2: Metrology of Materials in Flexible and Malleable Batteries

- Describe IV&V testing infrastructure and tools.
 - What tools and facilities can the team(s) access to validate properties of materials and interfaces relevant to the function of flexible or malleable battery systems?
- Describe key validation tests.
 - What key materials characterization tests are needed to understand the core performance and function of the flexible and malleable battery systems?

- What metrics and information can be obtained from these tests to support the validation of material composition as it relates to the electrochemical and mechanical functions?

Topic 3: Other Capabilities

- Describe other capabilities.
 - What other capabilities within the scope of this RFI are your organization able to provide? This can include modelling and simulation, manufacturing, and other support functions for flexible or malleable battery technologies.

SUBMISSION INSTRUCTIONS:

Responses to this RFI should be submitted no later than 5:00 p.m. ET on April 21, 2025.

Unclassified responses to this RFI should be submitted to DARPA-SN-25-64@darpa.mil. NO CLASSIFIED INFORMATION SHOULD BE SENT TO DARPA-SN-25-64@darpa.mil.

To the maximum extent possible, respondents should submit non-proprietary information. If proprietary information is submitted, it must be appropriately and specifically marked. It is the respondent's responsibility to clearly define to the Government what is considered proprietary data. Any proprietary information should be clearly labeled as "Proprietary." DARPA will disclose submission contents only for the purpose of review by DARPA staff, other Government agencies, or DARPA Support Contractors/SETAs.

NOTE: DARPA may conduct individual discussions with respondents as necessary to gain a full understanding of the technical capabilities described in the response. DARPA will contact respondents individually via e-mail.

FORMAT INSTRUCTIONS:

Responses to the RFI should be as concise as possible while also providing sufficient insight. Respondents should submit a single integrated response with page limits for each section indicated below. Effective responses that can be provided in fewer pages are encouraged. Format specifications for responses include 12-point font, single-spaced, single-sided, 8.5 by 11-inch page size with 1-inch margins in MS-Word or Adobe PDF format. Any submitted material that exceeds these limits will not be reviewed.

Responses should adhere to the following formatting instructions:

1. Cover page (1 page)
 - a. Title
 - b. Organization(s)
 - c. Respondent's technical and administrative points of contact (names, addresses, phone and fax numbers, and email addresses)
2. Technical Response to this RFI to include response to requested information about IV&V capabilities (4 pages maximum)
3. References (no limit)

This notice is not a request for solicitation, and awards will not be made through this notice. This information is subject to change in content, relative importance, or other ways without further notification.

DARPACONNECT:

Entities who have not worked with DARPA before are encouraged to learn more about DARPAConnect, an initiative established to facilitate collaboration between DARPA and potential performers. The DARPAConnect team offers customized support, resources, and guidance on how to prepare your ideas for high-impact conversations with DARPA program managers. Please visit [DARPAConnect.us](https://darpaconnect.us) to access a digital hub of online resources, including a curriculum for self-paced learning, personalized support, and in-person and virtual events. In addition to the self-paced online materials, the DARPAConnect team is able to schedule one-on-one conversations to discuss your specific ideas, questions, and paths to DARPA. You can use the contact form at [DARPAConnect.us](https://darpaconnect.us) or email the DARPAConnect team directly at darpaconnect@darpa.mil to request assistance.

ADMINISTRATIVE:

This announcement contains all information required to submit a response. No additional forms, kits, or other materials are needed. All administrative and technical questions should be directed to DARPA-SN-25-64@darpa.mil. Please refer to the Special Notice number (DARPA-SN-25-64) in all correspondence.

This RFI is issued solely for information and program planning purposes and does not constitute a formal solicitation for proposals or proposal abstracts; any so sent will be disregarded. In accordance with FAR 15.201(e), responses to this notice are not offers and cannot be accepted by the Government to form a binding contract. Submission of a response is strictly voluntary and is not required to propose to subsequent Announcements (if any) or Solicitations (if any) on this topic. DARPA will not provide reimbursement for costs incurred in responding to this RFI. Respondents are advised that DARPA is under no obligation to acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted under this RFI.